

Digital Material Passport

Manufacturer		Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Engineering Solutions Ltd. Tech Park Way 45 Cardiff CF14 5DU GB procurement@engisolutions.example.com	Order	PO-78902 · 2000 kg Pos. 10 · 2025-04-21
Subcustomer American Heavy Industries Inc. 5000 Industrial Blvd Building C Detroit, MI 48201 US materials@heavyind.example.com			Delivery	DN-56790 · 2000 kg Pos. 1 · 2025-05-13
		Goods Receiver Canadian Construction Corp 777 Construction Ave Toronto, ON M5H 2N2 CA logistics@canconstruct.example.ca		

Product

Product Name	Structural Steel S420N	Name (EN)	1.8902
Batch ID	H-10988-01	Shape	Plate - Width 2000 mm - Thickness 25 mm - Length 6000 mm
Surface Condition	Hot-rolled	Product Norm	EN 10025-3 (2019)
Weight	2000 kg		
Production Date	2025-05-10		
Country of Origin	DE		

Heat Treatment

Process: **Normalizing** - Lot: NORM-2024-0823-A12 - Furnace: NF-LINE-02 - Date: 2024-08-23

Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	920 C	60 min	Air	Air
Cooling	20 C			

Chemical Analysis

Heat Number **H-10988** - Melting Process **EAF+LF+VD** - Casting Method **ContinuousCasting** - Casting Date **2025-05-09** - Sample Location **Ladle**

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
C	%		0.2	0.16	P	%		0.025	0.016
Mn	%		1.6	1.48	S	%		0.015	0.01
Si	%		0.5	0.28	CEV	%		0.44	0.41

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength Test temperature: 20°C Specimen: 1/3R, L	Rm	558 / 562 / 560 MPa - Average 560 - Min 558 - Max 562 - Std Dev 2	520 MPa	680 MPa	EN ISO 6892-1	✓
Yield Strength Test temperature: 20°C, 3 specimens tested	ReH	442 / 445 / 448 MPa - Average 445 - Min 442 - Max 448 - Std Dev 3	420 MPa		EN ISO 6892-1	✓
Elongation after fracture Gauge length 5.65#S ₀ , 3 specimens tested	A	23 / 24 / 25 % - Average 24 - Min 23 - Max 25 - Std Dev 1	19 %		EN ISO 6892-1	✓
Reduction of Area¹	Z	60 / 62 / 64 % - Average 62 - Min 60 - Max 64 - Std Dev 2	50 %		EN ISO 6892-1	✓
Charpy V-notch Impact Energy² Specimen: Top, T-L	KV	56 / 58 / 60 J - Average 58 - Min 56 - Max 60 - Std Dev 2	40 J		EN ISO 148-1	✓
Charpy V-notch Impact Energy² Specimen: Bottom, T-L	KV	54 / 57 / 59 J - Average 56.7 - Min 54 - Max 59 - Std Dev 2.5	40 J		EN ISO 148-1	✓
Brinell Hardness Ball: 10mm, Force: 3000kg, 5 indentations	HBW	183 / 185 / 187 / 184 / 186 HBW - Average 185 - Min 183 - Max 187 - Std Dev 1.58	150 HBW	220 HBW	EN ISO 6506-1	✓

Vickers Hardness 10 kg load, 5 indentations	HV	192 / 195 / 198 / 194 / 196 HV10 - Average 195 - Min 192 - Max 198 - Std Dev 2.35	160 HV10	230 HV10	EN ISO 6507-1	✓
Rockwell Hardness	HR	18 HRC		22 HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa			EN ISO 6892-1	✓
Strain Hardening Exponent	n	0.18			ASTM E646	✓
Plastic Strain Ratio	r	1.2	1		EN ISO 10113	✓
0.2% Proof Strength¹	Rp0.2	428 / 430 / 432 MPa - Average 430 - Min 428 - Max 432 - Std Dev 2	400 MPa		EN ISO 6892-1	✓

¹ 3 specimens tested ² Test temperature: -20°C

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14